

# KAPIL RAVI RATHOD

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## Education

The Pennsylvania State University, University Park

August 2024 – May 2026 (Expected)

M.S. in Computer Science and Engineering

GPA - 3.89/4

National Institute of Technology, Warangal

July 2017 – May 2021

B.Tech in Computer Science and Engineering

CGPA - 7.98/10

## Experience

Systems and Network Security Lab (SyNSec)

September 2024 – Present

Student Researcher

State College, Pennsylvania

- **Conducting research** in identifying 5G network function security vulnerabilities, employing REST API fuzzing tools like RESTler to test and uncover potential security issues under the supervision of Dr. Syed Rafiul Hussain.

Deutsche Bank

July 2021 – July 2024

Associate Software Engineer, Associate Machine Learning Engineer

Pune, Maharashtra

- **Designed a cloud based RAG in Python** for Risk, Finance, Treasury, and Accounting, leveraging LLMs like Mistral, LLaMA2, and Gemini. **Saved \$3M annually**, supported **900 users** across **100+ teams**, and won the **India Excellence Award** for cost-effectiveness.
- **Developed** an LLM-based regulatory alerting service in Python, expected to **save the bank \$1M annually** and **700+ hours**, while ensuring compliance with central bank regulations across multiple countries.
- **Maintained and improved** a reporting application by addressing **OSS vulnerabilities**, performing **Tomcat server upgrades** and **UI enhancements**. Awarded the **recognition award** for project contributions.
- **Automated** the HR CV review process by developing a Python-based CV scanning tool, **reducing manual effort by 80%** and serving **150+ hiring managers**, deployed using a **Jenkins CI/CD** pipeline to an **OpenShift** cluster.
- **Eliminated** manual efforts in SAP pre-deployment checks through **Python** and **Unix automation** scripts.
- **Enhanced** a Python based document translation service, **improving BLEU scores by 12%** across all internal webpages by replacing the existing Opus-MT, M2M100 models with the WMT21 model.

Semut.io

April 2020 – July 2020

Cloud Native Engineering Intern

Bengaluru, Karnataka

- **Contributed** to the development of a Golang based application manager to provide Elasticsearch as a service.

Here Technologies

May 2019 – July 2019

Software Engineering Intern

Mumbai, Maharashtra

- **Designed** a spell-check system **saving 100+ hours** for the map creation team.
- **Developed** a proof of concept for stitching Here vehicle street images, enhancing perception for autonomous vehicles.

## Personal Projects

Northropp Grumman Object Detection Hackathon | Python, Deep Learning

September 2024

- **Designed** a real-time object detection system using YOLOv7 and Grounding DINO models. Secured second place and won \$1500 prize in the competition.

Composed Image Retrieval on a Fashion Dataset | Python, Multi Modal Image Retrieval

September 2024

- **Developed** a CIR model using the CLIP vision-language model to retrieve styled clothing images based on a reference image and text modifier, enhancing visual and textual integration on a specialized dataset.

InfraAlert | Python, React, GCP, Deep Learning

July 2023

- **Developed** a real-time cloud-based system in Python to predict system parameters and downtime, based on RAM, CPU, Disk, and NetPacket usage using LSTM and stacked CNN-LSTM models, leveraging data from a deployed application.

Satya - Smart India Hackathon(SIH) 2020 | Django

August 2020

- **Built** an online platform enabling users to report instances of corruption in public sector and government offices, securing the **National Winner** title.

## Publications

- **K.R. Rathod, S. Bansal, Y.H.C. Kandpal and M. Bisi "Learning Automata-based Feature Selection for Duplex Output Software Effort Estimation Model"**, 12th ICCCNT, Kharagpur, India, 2021. [Link]

## Technical Skills

Languages: Python, Java, C, C++, Golang, R, Javascript, Typescript, MATLAB

Relational and NoSQL Databases: MySQL, Oracle, MongoDB

Vector Databases: Qdrant, Milvus

Cloud: Google Cloud Platform [Link]

Technologies/Frameworks: Django, Angular, Springboot, Unix, Docker, Kubernetes, Jenkins, Git, Langchain, OpenShift